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Study on Enhancing Recovery by Multi-Scale Geomechanics Reconstruction and Combined Displacement-Imbibition in Unconventional Resources

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Abstract

In the late stage of development in the X77 Block, reservoir stress depletion has led to rapid productivity decline, requiring an adjustment of the development strategy. A coupled seepage-geomechanics embedded fracture-flow model was established to simulate multi-field interactions, revealing the evolution of pressure, flow, and stress fields after long-term water injection. Results indicate that pressure diffusion is limited, preventing the formation of an effective displacement system, and that changes in minimum horizontal stress exceed those in maximum horizontal stress. A multi-field reconstruction and enhanced oil recovery approach was implemented by re-establishing an artificial fracture-network system through large-scale fracturing, coupled with full-cycle energy supplementation methods-pre-frac energy charging, intra-frac energy boosting, shut-in energy storage, and displacement-drainage integration-to address crossflow and insufficient energy replenishment. Multi-well synergy was used to enhance imbibition and integrate displacement with drainage under a complex fracture network, shifting the strategy from avoiding fractures to utilizing them, thereby improving microscopic sweep and displacement efficiency. Considering gas supply, recovery degree, and economic benefits, four development scenarios were compared, and Scenario 3: CO₂ gravity flooding-was recommended as optimal, involving 3 injectors and 9 producers in S6, and 6 injectors and 33 producers in S7+S8, enabling recovery of 299.6 million tonnes of geological reserves. This provides strong technical support for long-term stable production in the X77 Block.

Fracture Flow Model Coupling Seepage and Geomechanics

To characterize the changes in the stress field, seepage field, and energy field of tight reservoirs after long-term water injection, a fracture flow model coupling seepage and geomechanics was established. This model considers the flow and phase transition processes of multi-component fluids, the fluid-solid coupling process of reservoir media under geomechanical effects, and the mechanical behavior of fractures.

To ensure computational efficiency, a simulation framework based on the fixed-stress iterative coupling solution method was designed, enabling the seamless integration of the geomechanical model, seepage model, and fracture model. Compared to traditional models, the proposed method quantifies the influence of in-situ stress and fractures on the flow in tight reservoirs, providing a quantitative basis for evaluating and reconstructing the stress field, seepage field, and energy field.

Reservoir Seepage Equation Coupled with Geomechanics

Assumptions: The fluid is slightly compressible. The rock undergoes only linear elastic deformation. The mechanical process is quasi-static. The rock deformation satisfies the small deformation hypothesis. The extraction process is isothermal with no chemical reactions. The geomechanical equation of the reservoir can be established as:

$$\begin{cases} \nabla \cdot (C_{dr} : \varepsilon - b p_E \delta) + [\phi \rho + (1 - \phi) \rho_s] g = 0 \\ \varepsilon = \frac{1}{2} [\nabla u + (\nabla u)^T] \end{cases} \quad (1)$$

$$P_E = S_W P_W + S_o P_o + S_g P_g \text{ and } P_f = S_W P_W + S_o P_o + S_g P_g$$

Based on the mass conservation theory and Darcy's law, the flow control equations for carbon component *i* and water, considering rock deformation, are:

$$\begin{aligned} & \frac{\partial [(x_i S_o \rho_o + y_i S_g \rho_g) \phi]}{\partial t} + \frac{\partial [(x_i S_o \rho_o + y_i S_g \rho_g) \phi] \varepsilon_v}{\partial t} = \\ & \nabla \cdot \left[x_i \frac{\rho_o K_{ro}}{\mu_o} K(\nabla p_o - \rho_o g) + y_i \frac{\rho_g K_{rg}}{\mu_g} K(\nabla p_o - \rho_o g) \right] - q_i \\ & (i = 1, 2, \dots, n_c) \end{aligned} \quad (2)$$

$$\frac{\partial (S_w \rho_w \phi)}{\partial t} + \frac{\partial (S_w \rho_w \phi) \varepsilon_v}{\partial t} = \nabla \cdot \left[\frac{\rho_w K_{rw}}{\mu_w} K(\nabla p_w - \rho_w g) \right] - q_w \quad (3)$$

The fugacity balance equation for each carbon component is constructed to constrain variables:

$$f_{L,i} - f_{V,i} = 0 \quad (4)$$

Additionally, the mole fraction and saturation must satisfy the following linear constraints:

$$\sum_{i=1}^{n_c} x_i - 1 = 0 \quad (5)$$

$$\sum_{i=1}^{n_c} y_i - 1 = 0 \quad (6)$$

$$S_w + S_g + S_o = 1 \quad (7)$$

Fracture Model

The fracture model includes the calculation models for fracture deformation and fracture constitutive relationships. The displacement discontinuity method (DDM) is used to calculate the true deformation of fractures, where the solving variables are set as the normal displacement (D_n) and tangential displacement (D_s). In solving the problem, all fractures in space are first discretized into grids. For a two-dimensional problem, they are divided into a series of line segments. The normal and tangential displacements of each element are defined based on the local coordinate system $O-xy$:

$$\begin{cases} D_s = u_x(x, 0_-) - u_x(x, 0_+) \\ D_n = u_y(x, 0_-) - u_y(x, 0_+) \end{cases} \quad (8)$$

Under this definition, the displacement discontinuity generated by the element at any point in space is expressed as:

$$\begin{cases} u_x = [2(1-\nu)f_y - yf_{xx}] + [- (1-2\nu)g_x - yg_{xy}] \\ u_y = [(1-2\nu)f_x - yf_{xy}] + [2(1-\nu)g_y - yg_{yy}] \\ \sigma_{xx} = 2G(2f_{xy} + yf_{xyy}) + 2G(g_{yy} + yg_{yyy}) \\ \sigma_{yy} = 2G(-yf_{xyy}) + 2G(g_{yy} - yg_{yyy}) \\ \tau_{xy} = 2G(2f_{yy} + yf_{yyy}) + 2G(-yg_{xyy}) \end{cases} \quad (9)$$

Within the fracture element, the local conservation of fluid is controlled by the mass balance equation, which can be expressed as:

$$\frac{\partial(qw)}{\partial s} + \frac{\partial(w)}{\partial t} + q_L = 0 \quad q_L = \frac{2C_L}{\sqrt{t-t_0}} \quad (10)$$

$$q_L = \frac{2C_L}{\sqrt{t-t_0}} \quad (11)$$

Additionally, the fluid must satisfy the global volume conservation law:

$$\int_0^t Q(t)dt = \int_{H_f} \int_0^{L(t)} wdsdh + \int_{H_f} \int_0^{L(t)} \int_0^t q_L dt ds dh \quad (12)$$

The constitutive model of the fracture is constructed using the Barton-Bandis model, with the compliance model describing the constitutive relationship. The compliance model is a diagonal matrix composed of normal stiffness (Kn) and tangential stiffness (Ks):

$$C = \text{diag}\left(\frac{1}{K_n}, \frac{1}{K_s}, \frac{1}{K_s}\right) \quad (13)$$

Spatial Discretization and Solution Method

The Embedded Discrete Fracture Model (EDFM) is employed to simulate the influence of fracture geometry on flow. In geomechanical problems, the EDFM geometry is equivalently treated using the Oda method. For fluid flow problems, EDFM embeds fractures into the matrix grid as additional grids, and the flow connectivity of fractures is handled by introducing non-neighboring connections (NNC).

In this study, the Karmi-Fard method is used to calculate the transmissibility.

$$T = KA/d_n \quad (14)$$

For the first type of NNC, the characteristic area (An) is the intersection area between the fracture and the grid, the permeability (K) is the harmonic average of the matrix permeability and the fracture permeability, and the characteristic distance (dn) is a function of the unit control volume and the unit volume:

$$d_n = \frac{\int_{V_c} d dV}{V_c} \quad (15)$$

For the second and third types of NNC, the transmissibility is the harmonic average of the two fracture segments, where the transmissibility of a single fracture segment can be expressed as a function of the fracture width, the length of the fracture intersection line, and the distance between the centers of the fracture segments:

$$T_f = \frac{K_f w_f L_{\text{int}}}{d_f} \quad (16)$$

The fixed-stress iterative coupling method is used to perform the coupled geomechanical-seepage simulation, and the explicit coupling method is applied to simulate the fracture system. During the solution process, the conservation requirements of the subproblem variables and the numerical stability requirements are fully considered. The finite element method is used to discretize the geomechanical equations, the finite difference method is used to discretize the fluid flow equations, and the displacement discontinuity method is used to discretize the fracture propagation equations. Thanks to the advantages of the embedded discrete fracture model, there is no need to re-mesh the background grid during the solution process. Only the fracture geometry needs to be updated, and the connectivity between the fractures and the matrix grid is recalculated, significantly reducing computational costs.

In the initialization phase, for the first-time solution, the gravity equilibrium method is used to initialize the fluid pressure field, and then the initial in-situ stress field is calculated based on the pressure field. For restart problems, the simulation results from the previous time step are directly substituted into the solution. The entire fluid-solid coupling process is completed within the same time step. After solving the fluid-solid coupling process, the fracture geometry and parameters are updated based on the current stress state of the grid, and it is determined whether the fracture grid needs to be updated for the current time step.

3D Modeling

The oil-bearing formation is characterized by fining-upward depositional cycles, with distinct sedimentary structures developing at different parts of each cycle. In the lower cycle, common features include channel-base scour surfaces and parallel bedding, often accompanied by graded bedding, cross-bedding, trough cross-bedding, and massive structures. The middle to upper cycle typically exhibits small- to micro-scale cross-bedding, trough cross-bedding, ripple lamination, and horizontal bedding, with occasional plant debris fossils, coal streaks, and bitumen in the sandstone. Abundant channel-base scour surfaces indicate a predominantly fluvial depositional environment, while parallel bedding reflects high-energy conditions often associated with scour surfaces. Graded bedding suggests gravity sorting, producing the characteristic fining-upward sequence. Massive bedding reflects rapid deposition under abruptly reduced flow energy. Cross-bedding and trough cross-bedding in the middle to upper sections indicate traction current deposition under moderate flow. Fine-grained ripple laminations near the cycle top suggest lacustrine influence under very low energy, and horizontal bedding typically reflects deposition in stagnant water. Plant debris and coal streaks are interpreted as material transported by river channels. The surface model defines the three-dimensional distribution of stratigraphic interfaces, derived from well correlation data and seismic horizon interpretations. Well-based stratigraphic data provide the foundation, while seismic depth-domain structural horizons serve as trend surfaces, ensuring alignment at well locations and consistency with seismic trends between wells. Fault integration produces a structural model that captures the present distribution of the target interval and the post-deformation configuration of the original sediments. Using existing well stratigraphy and constraining with seismic-derived thicknesses of the upper and lower reservoir sections, a stratigraphic model for both intervals was constructed (Figure 1).

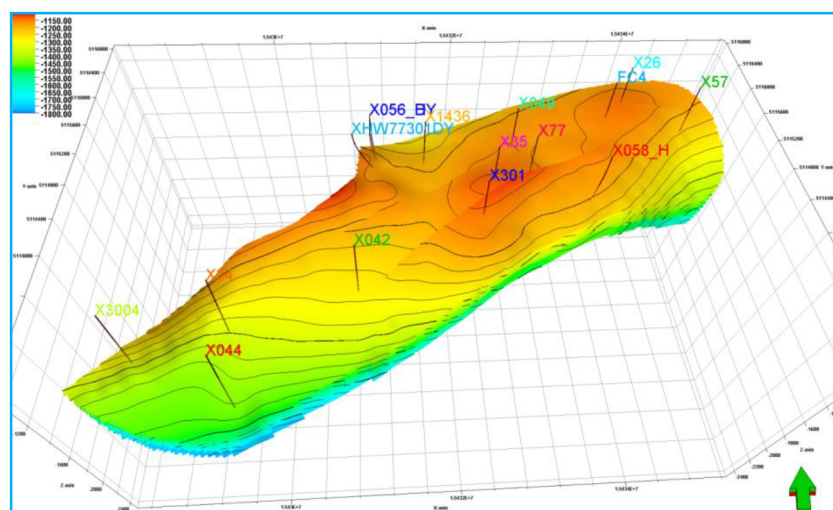


Figure 1—X77 Block Layering Model

Reservoir 3D modeling is performed within the established structural framework by gridding the stratigraphic model in three dimensions, dividing the geological body into a series of grids along the X, Y, and Z axes. The grid system consists of horizontal and vertical components, with horizontal grid spacing set at 30×30 m based on the work area size, well density, and modeling accuracy requirements. Vertical grid design considers both accuracy and stratigraphic stacking relationships, with the reservoir divided into four vertical zones: T2k11, T2k12-1, and T2k12-2, each subdivided into 100 layers, and T2k13 into 150 layers. The final model comprises a total of 8,706,104 grid cells (Figure 2).

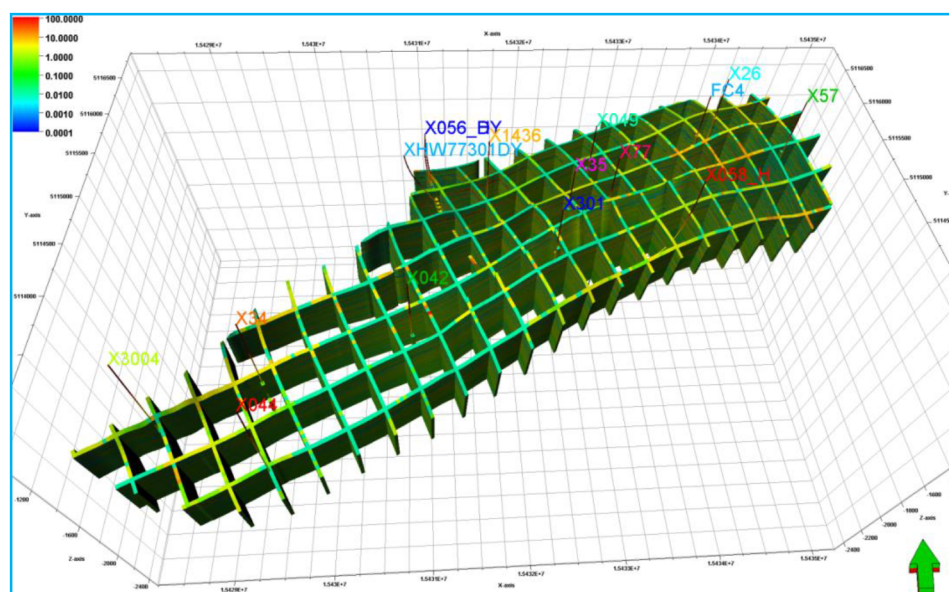


Figure 2—X77 Block Grid model

The reservoir facies model enables quantitative and detailed characterization of both vertical and lateral heterogeneity, providing an accurate representation of facies distribution, which forms the foundation for constructing high-resolution 3D reservoir property models. 3D facies modeling generally follows a point-to-surface-to-volume approach, starting with one-dimensional vertical models at each well, then developing the reservoir facies model based on the established depositional understanding. Modeling methods include deterministic and stochastic approaches: deterministic modeling derives a single, unique facies distribution

between wells based on known data, while stochastic modeling applies probabilistic methods to interpolate unknown interwell areas, generating equally probable realizations of reservoir distribution.

Resistivity effectively differentiates mudstone, fine sandstone, argillaceous conglomerate (poor reservoirs), and conglomerate (favorable reservoirs). Favorable reservoirs exhibit high acoustic impedance. By integrating well and seismic data, an acoustic impedance inversion volume was generated, with conglomerate (favorable reservoir) characterized by impedance values $\geq 10,500 \text{ g/cm}^3 \cdot \text{m/s}$. Due to limited well data in the study area, inter-well reservoir distribution exhibits high randomness. A stochastic modeling approach was applied to generate an ensemble of equiprobable 3D reservoir models, enabling uncertainty assessment and better understanding of reservoir variability between wells. To minimize model uncertainty, deterministic information was incorporated to constrain the stochastic process. The reservoir facies model was constructed using a facies-controlled sequential indicator simulation method that combines deterministic and stochastic modeling. Before constructing the 3D reservoir model, variogram analysis was conducted to characterize the spatial correlation of variables with distance. By understanding key variogram parameters, particularly the range, geological insights can exert stronger control over the modeling process. Variogram analysis was performed for each lithofacies parameter to determine their spatial continuity, which served as the basis for property distribution between wells. The results indicate a major range of 1,500–2,000 m, a minor range of 800–1,000 m, and a vertical range of approximately 10 m, reflecting anisotropy in reservoir properties. To accurately reflect spatial lithology, high-quality 3D seismic data constraints were fully utilized. The lithology model was constrained by impedance inversion volumes (Figure 3).

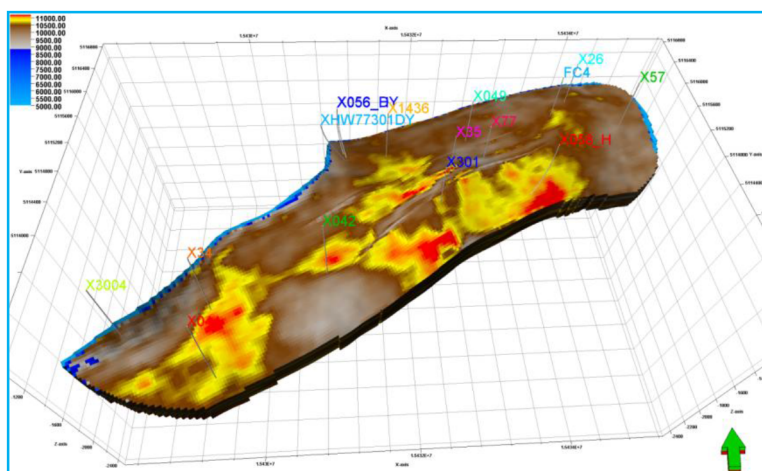


Figure 3—X77 Block Impedance Model

Based on single-well lithology statistics and using seismic inversion volumes as inter-well constraints, a lithofacies model of the study area was constructed using the sequential indicator simulation method. This approach accurately characterizes the spatial distribution of different lithologies in three dimensions, providing constraints for subsequent property modeling. Inter-well lithofacies predictions are controlled by inversion-derived probability model parameters, improving reservoir facies prediction accuracy. Favorable reservoirs (conglomerates) are primarily distributed in structurally high positions and the southwest area, while most of the northeast is dominated by poor-quality reservoirs (fine sandstones).

Scheme Design and Optimization

Three gas injection gravity drainage schemes were designed for comparative evaluation (Table 1).

1. Base Plan: Horizontal Well Depletion Development

Recoverable geological reserves are estimated at 2.996×10^6 t, targeting the S6, S7, and S8 formations. The development strategy adopts depletion drive with a row-pattern well layout comprising 51 producing wells (Figure 3.3–39). Under this mode, annual oil production is maintained at 2.0×10^5 t for four years. Over a 25-year evaluation period, cumulative oil production reaches 2.24×10^6 t, with cumulative fluid production of 4.026×10^6 t, corresponding to a recovery factor of 7.47%.

2. Scheme 1: CO₂ Gravity Drainage with Pattern Wells

Recoverable geological reserves total 2.996×10^6 t, targeting the S6, S7, and S8 formations. Development employs CO₂ gravity drainage with a row-pattern well configuration: S6 contains 3 injectors and 9 producers, while S7+S8 contain 6 injectors and 33 producers. Injection parameters include a daily CO₂ rate of 2,300 t (1.20×10^6 m³) and cumulative CO₂ injection of 1.725×10^7 t (8.71×10^8 m³). Annual oil production is maintained at 5.0×10^5 t for eight years. Over the 25-year evaluation period, cumulative oil production reaches 1.0232×10^7 t, achieving a recovery factor of 34.15%, representing a 26.7 percentage point increase over the base case.

3. Scheme 2: N₂ Gravity Drainage with Combined Well Pattern

Recoverable geological reserves total 2.996×10^6 t, targeting the S6, S7, and S8 formations. Development applies nitrogen gravity drainage with a combined well pattern: S6 includes 3 injectors and 9 producers, while S7+S8 include 6 injectors and 33 producers. Injection parameters consist of a nitrogen injection rate of $1.0\text{--}0.4 \times 10^6$ m³/d and a cumulative injection of 3.49×10^9 m³. Annual oil production is maintained at 5.0×10^5 t for six years. Over the 25-year evaluation period, cumulative oil production reaches 9.185×10^6 t, achieving a recovery factor of 30.6%, which is 23.2 percentage points higher than the base case.

4. Scheme 3: Mixed Gas Gravity Drainage with Combined Well Pattern

Recoverable geological reserves total 2.996×10^6 t, targeting the S6, S7, and S8 formations. Development adopts CO₂ gravity drainage with a combined well pattern: S6 includes 3 injectors and 9 producers, while S7+S8 comprise 6 injectors and 33 producers (Figure 4 and Figure 5).

Table 1—Xia77 Reservoir Gas Injection Gravity Drainage Scheme Designs

Scheme	Base Plan	Scheme 1	Scheme 2	Scheme 3
Development	Old wells with current regime	New horizontal wells, S6 & S7	New horizontal wells, S6 & S7	New horizontal wells, S6 & S7
Method	depletion	CO ₂ gravity drainage	N ₂ gravity drainage	Mixed gas gravity drainage
Wells (inject/produce)	51 producers	7 injectors (S6 3 inj, S7+ 4 inj), 42 producers	7 injectors (same), 42 producers	7 injectors (same), 42 producers

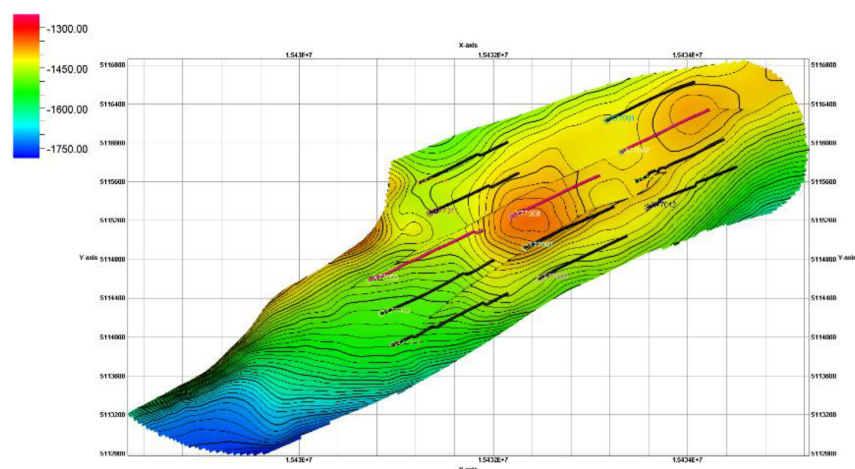


Figure 4—X77 well deployment plans for S6 layer

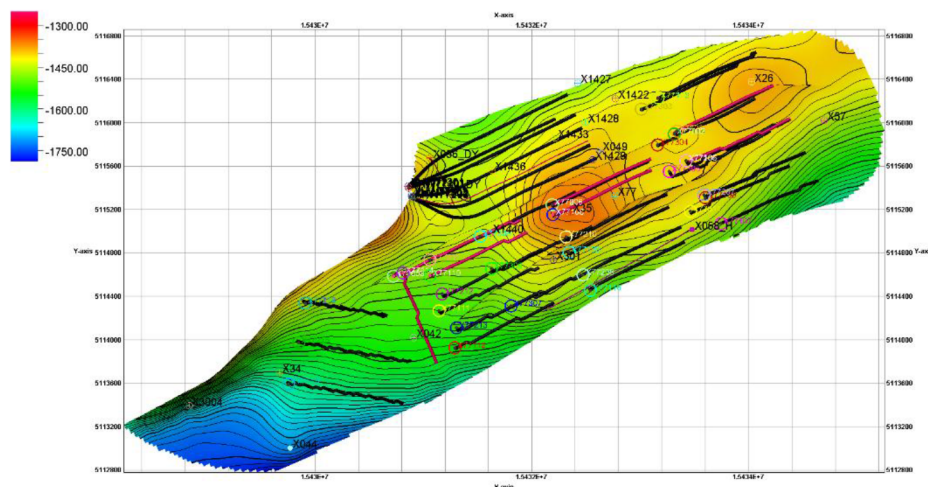


Figure 5—X77 well deployment plans for S7+S8 layers

Injection–production parameters are based on a mixed gas medium (20% CO₂ + 80% N₂) over a 4-year injection period. Daily injection rates are 1.20×10^6 m³ CO₂ (cumulative 1.58×10^9 m³, 0.2 PV) and 1.00 – 0.40×10^8 m³ N₂ (cumulative 3.76×10^9 m³). Oil production is maintained at 5.0×10^5 t/year for 7 years. Over a 25-year evaluation period, cumulative oil output reaches 9.572×10^6 t, with a recovery factor of 31.9%, representing a 24.5 percentage point increase over the base case (Figure 6 and Figure 7).

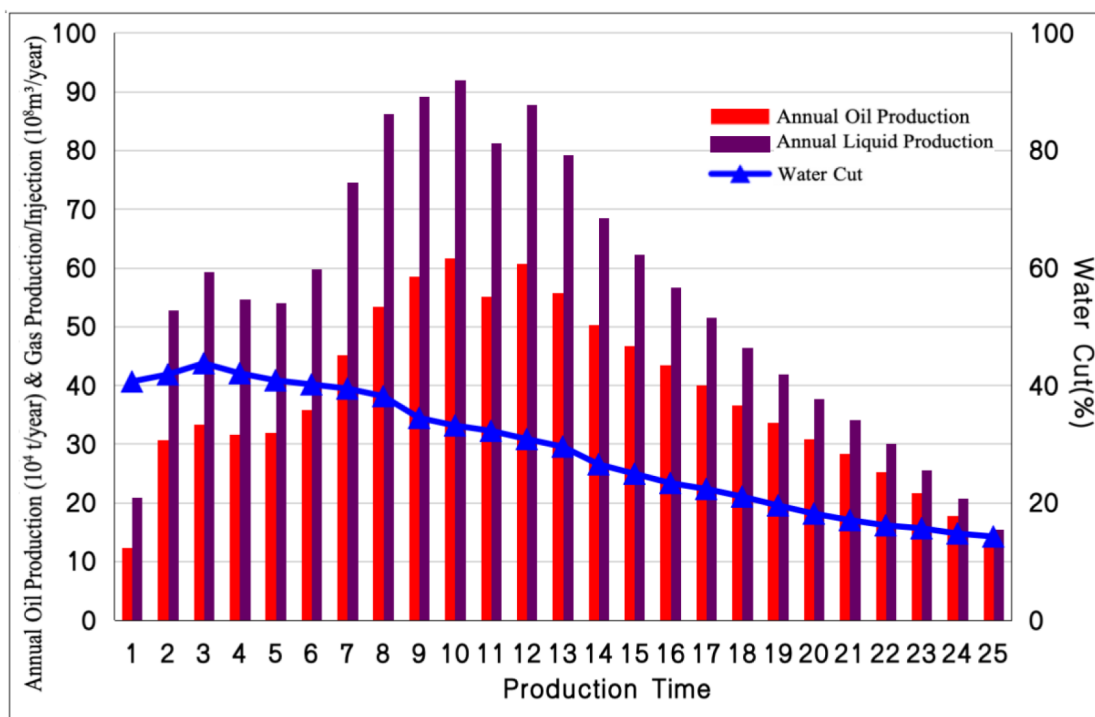


Figure 6—Comparison of Development Indicators for Scheme 3 (Oil Production)

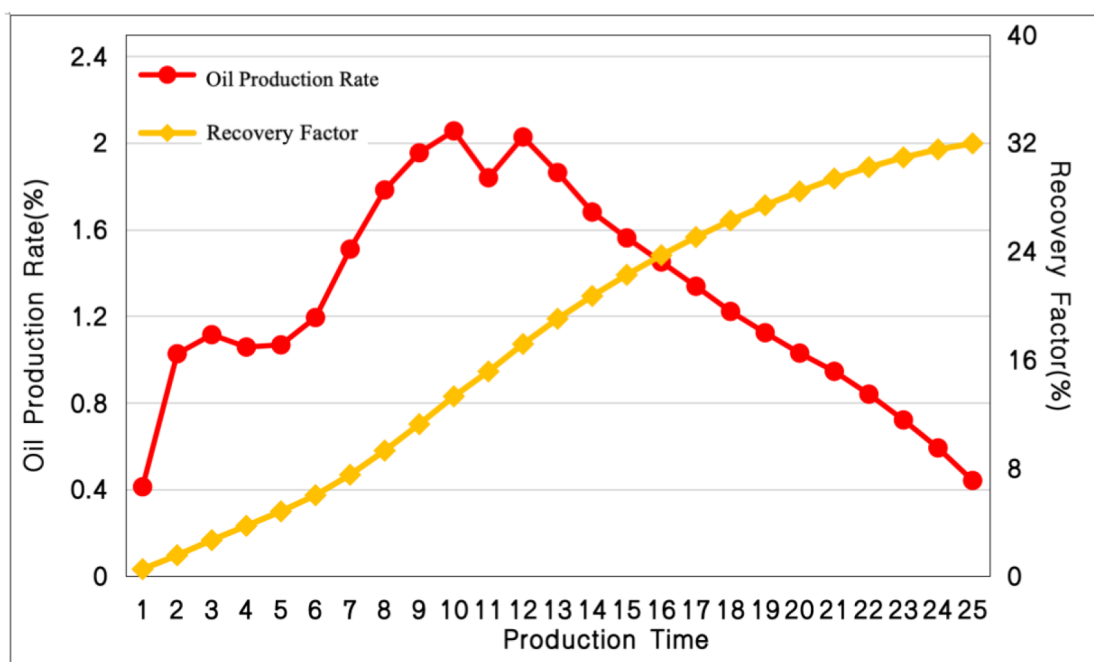


Figure 7—Comparison of Development Indicators for Scheme 3 (Oil Rate and Recovery Factor)

Scheme 3 is recommended considering gas availability, recovery, and economics (Table 2).

Table 2—Key Development Indicators Comparison

Scheme	Total wells	Producers	Gas Injectors	Water Injectors	Plateau Oil Rate (10 ⁴ t/a)	Plateau Duration	Total Requirement (10 ⁸ m ³)	Gas (10 ⁸ m ³)	Cumulative Oil Production (10 ⁴ t)	Recovery Factor (%)	Incremental Recovery over Base (%)
Scheme 1 (CO ₂ Injection, Pattern Well Layout)	51	42	9	/	50	8	87.1		1023.2	34.15	26.7
Scheme 2 (N ₂ Injection, Combined Well Layout)	51	42	9	/	50	6	34.9		918.5	30.6	23.2
Scheme 3 (Mixed Gas Injection, Combined Well Layout)	51	42	9	/	50	7	53.4		957.2	31.9	24.5
Base Case	51	51	/	/	20	4	/		224	7.47	/

Conclusions

By integrating well–seismic geological body identification and prediction techniques, a high-resolution 3D geological model of the X77 block was reconstructed. Based on single-well lithofacies data and constrained by seismic inversion, lithofacies and property models were built to accurately depict the vertical migration

of the gas–liquid interface during gas gravity drainage. Vertical grid resolution was optimized, resulting in a fine 3D model of $15 \times 15 \times 0.5$ m (8,538,750 cells).

Benchmarking domestic and international cases, a geology–engineering–gas injection integrated approach was applied to evaluate the feasibility of gas gravity drainage, optimize key injection–production parameters, and design dynamic monitoring schemes. Considering the spatial distribution of induced fracture networks, a 3D staggered horizontal well pattern of 51 wells was deployed, with an innovative stepwise staggered injection strategy. Reservoir engineering methods and compositional simulation were used to optimize parameters such as injection–production ratio, injection volume, rate, and timing. Comparative analysis of three reservoir engineering schemes recommended a combined-well mixed-gas gravity drainage strategy, achieving 95.72 Mt cumulative oil production and a 31.9% recovery factor—24.5% higher than the base case. Following a phased implementation plan, pilot areas were selected with optimized production targets and monitoring programs.

After long-term waterflooding, the reservoir exhibited limited pressure propagation and low energy transfer efficiency, with greater variation in minimum horizontal stress than maximum horizontal stress, and more pronounced stress changes in injectors than producers. A multi-field reconstruction and displacement integration approach was proposed, involving conversion of injectors to producers, large-scale fracturing, pre-fracture energy supplementation, optimized fluid placement for maximum fracture extension, and post-fracture shut-in to expand the energy field and enhance oil–water displacement. Field application confirmed that this method significantly improves recovery, providing strong technical support for efficient reservoir development.

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Nomenclature

- A_n – Characteristic area, m²;
- b – Biot coefficient;
- C – Compliance model matrix;
- C_{dr} – Fourth-order tensor of elastic drained bulk modulus, Pa;
- CL – Carter leak-off coefficient, m/s^{1/2};
- d – Distance between adjacent grid centers, m;
- df – Distance between adjacent fracture element centers, m;
- dn – Characteristic distance, m;
- D_n, D_s – Normal and tangential displacement, m;
- $f_{L,i}, f_{V,i}$ – Fugacity of component i in liquid and gas phases, Pa;
- f_x, f_y – First-order derivatives of the constant Green's function with respect to x and y , m;
- f_{xx}, f_{xy}, f_{yy} – Second-order partial derivatives of the constant Green's function with respect to xx , xy , and yy , dimensionless;
- f_{xyy} – First-order derivative of the constant Green's function with respect to x and second-order derivative with respect to y , m⁻¹;

- f_{yyy} – Third-order derivative of the constant Green's function with respect to y , m^{-1} ;
 g – Gravitational acceleration, m/s^2 ;
 $\mathbf{g}$ – Gravitational acceleration vector, m/s^2 ;
 g_x, g_y – First-order derivatives of the linear Green's function with respect to x and y , m ;
 g_{xy}, g_{yy} – Second-order partial derivatives of the linear Green's function with respect to xy and yy , dimensionless;
 g_{xyy} – First-order derivative of the linear Green's function with respect to x and second-order derivative with respect to y , m^{-1} ;
 g_{yyy} – Third-order derivative of the linear Green's function with respect to y , m^{-1} ;
 G – Shear modulus of elastic medium, Pa;
 h – Height, m;
 hL – Height of leak-off zone, m;
 Hfl – Fluid height within the fracture, m;
 i – Carbon component index;
 K – Absolute permeability, m^2 ;
 K_f – Fracture permeability, m^2 ;
 K_n, K_s – Normal and tangential stiffness, N/m;
 K_{rw}, K_{rg}, K_{ro} – Relative permeability of water, gas, and oil phases, %;
 L_{int} – Length of fracture intersection line, m;
 n_c – Number of carbon components;
 p_E – Equivalent fluid pressure, Pa;
 p_w, p_g, p_o – Pressure of water, gas, and oil phases, Pa;
 q – Fluid flow velocity in the fracture, m/s;
 q_i – Mass flow rate of carbon component i per unit volume, $kg/(m^3 \cdot s)$;
 q_L – Leak-off rate, m/s;
 q_w – Mass flow rate of water phase per unit volume, $kg/(m^3 \cdot s)$;
 Q – Fluid flow rate, m^3/s ;
 s – Fluid flow distance, m;
 S_w, S_g, S_o – Saturation of water, gas, and oil phases, %;
 t – Time, s;
 t_0 – Time when the fracture element first contacts the fracturing fluid, s;
 T – Transmissibility, m^3 ;
 T_f – Transmissibility of a single fracture segment, m^3 ;
 $u_x(x, 0^-)$ – Displacement along the negative direction of the fracture strike, m;
 $u_x(x, 0^+)$ – Displacement along the positive direction of the fracture strike, m;
 $u_y(x, 0^-)$ – Displacement perpendicular to the negative direction of the fracture strike, m;
 $u_y(x, 0^+)$ – Displacement perpendicular to the positive direction of the fracture strike, m;
 $\mathbf{u}$ – Displacement vector, m;
 V – Volume, m^3 ;
 V_c – Control volume of the element, m^3 ;
 w – Average width, m;
 w_f – Fracture width, m;
 x, y – Cartesian coordinates, m;
 x_i, y_i – Mole fraction of carbon component i in oil and gas phases, %;
 δ – Second-order Kronecker delta;
 ϵ – Strain tensor, dimensionless;
 ϵ_v – Volumetric strain, defined as the trace of the strain tensor ϵ in the flow equation;

- μ_w, μ_g, μ_o – Viscosity of water, gas, and oil phases, Pa·s;
 ρ_f – Total fluid density, kg/m³;
 ρ_s – Rock density, kg/m³;
 ρ_w, ρ_o, ρ_g – Density of water, gas, and oil phases, kg/m³;
 σ_{xx}, σ_{yy} – Normal stress components in xx and yy directions, Pa;
 τ_{xy} – Shear stress component in xy direction, Pa;
 ν – Poisson's ratio;
 ϕ – True porosity, %.

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